HAMZA MAJID

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• LinkedIn • GitHub • Portfolio

SUMMARY

AI/ML Engineer combining 2+ years of production software engineering in regulated financial environments with active AI/ML research at TU Berlin. Currently pursuing an MSc in Artificial Intelligence at Brandenburg University of Technology. Recent project work includes a comparative study of Graph Neural Networks (GraphSAGE) and Transformers for road network trajectory prediction, alongside time-series forecasting with LSTMs and NLP pipeline development. Skilled in Python, deep learning, graph neural networks, and scalable system design. Seeking thesis and internship opportunities in AI engineering, autonomous systems, or intelligent network research in Germany.

WORK EXPERIENCE

Working Student | Technische Universität Berlin | Berlin, Deutschland | May 2026 – Present

- Deployed a software defined private 5G O-RAN testbed using OpenAirInterface and FlexRIC for AI-driven network management and XAI validation.
- Built a real-time KPI pipeline (E2 → SQLite → InfluxDB → Grafana) capturing per-UE metrics for model training.
- Integrated RC service model and designed data export for RL-based resource allocation and XAI research.
- Fixed critical E2 integration by replacing O-RAN SC RIC with FlexRIC, enabling stable end-to-end data flow.
- Automated testbed setup and operations via Python and shell scripts for reproducibility.

Working Student | Technische Universität Berlin | Berlin, Deutschland | August 2025 – April 2026

- Optimized the Autonomous Reservoir Computing pipeline, increasing time-series prediction accuracy by 65%.
- Enhanced the PyReco open-source library (reservoir computing/ESN) used by AI/ML researchers globally.
- Developed an interactive neural network visualization tool, improving reservoir structure understanding for new users.
- Implemented a robust error-handling framework using validator and decorator patterns, improving code reliability and user feedback.

Software Engineer | Bank Al Habib | Karachi, Pakistan | August 2022 – April 2025 |
(incl. Trainee, Aug 2022 – Mar 2023)

- Developed and maintained 9+ data-driven microservices using Java and Quarkus on Linux for core banking operations.
- Improved system performance by reducing execution time by 40% through algorithm and code optimization.
- Implemented Apache ActiveMQ based event-driven architecture, reducing inter-service latency by 95%.
- Designed IBM Db2 schemas and reporting solutions for financial analytics and risk monitoring.
- Contributed to agile development and code reviews, ensuring high-quality, production-ready systems.

EDUCATION

Master of Science in Artificial Intelligence | April 2025 – Present
Brandenburg University of Technology, Cottbus

Bachelor of Science in Computer Science | September 2018 – May 2022
Institute of Business Management, Karachi

SKILLS

- **Programming Languages:** Java, Python, C, MATLAB, SQL
- **Frameworks & Technologies:** Quarkus, Apache ActiveMQ, REST APIs, Micro-services Architecture, Spark, Databricks, Grafana, FlexRIC, OpenAirInterface
- **Database & Storage:** MySQL, PostgreSQL, DB2, Optimization, Data Warehousing, Influx DB

- **AI & Machine Learning:** Supervised & Unsupervised Learning, Neural Networks, Graph Neural Networks, Transformers, Reservoir Computing, Scikit-learn, PyTorch, OpenCV, LLM, NLP, Reinforcement Learning, Explainable AI (XAI)
- **Messaging & Event-Driven Systems:** Apache ActiveMQ, JMS Queues, Transaction Management
- **Quantitative & Financial Analysis**: Time-Series Analysis, Statistical Modeling, Regression, Forecasting, Optimization, Financial Data Analysis, Risk Metrics, Backtesting, Numerical Methods
- **Version Control & DevOps:** Git, GitHub, Jira, CI/CD, Docker
- **Tools & IDEs:** IntelliJ, VS Code, Eclipse; APIs: Postman/Insomnia
- **Languages**: English-C1, Urdu-Native, German-A2/B1

PROJECTS

Graph-Based Trajectory Prediction using GNNs and Transformers

- Built a road network graph from OpenStreetMap data (~2K nodes, ~5K edges) and generated movement trajectories via random walks.
- Implemented GraphSAGE for spatial relationships and a Transformer for sequential patterns, with full training pipelines.
- Compared models on next-node prediction accuracy and cross-entropy loss: Transformers outperformed GNNs on sequential trajectory modeling; GNNs captured spatial road structure more effectively findings directly applicable to autonomous driving architecture decisions.

Financial Time-Series Forecasting (Stock Market Prediction)

- Built a multi-ticker prediction pipeline for SPY, QQQ, and equities, comparing Logistic Regression, XGBoost, and LSTM using 55+ leakage-free technical and market-regime features.
- Implemented expanding-window walk-forward validation with purge/embargo gaps and majority-class baseline (AUC ~0.50–0.54), plus back testing with transaction costs, Sharpe, drawdown, and position-sizing overlays.
- Developed an interactive Streamlit dashboard for per-ticker training, walk-forward evaluation, back testing, and predictions, with optional LSTM support and transparent performance reporting.

Recall — Multi-Document RAG Pipeline

- Built a multi-document RAG pipeline with vector database retrieval, comparing chunking strategies and embedding models on retrieval quality. Evaluated with recall @ k metrics against a held-out question set.